



Finish Masking

Procedure SKF_610

Revision: C
Date: 5/18/22

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1. SCOPE

1.1. Purpose

This procedure describes the application of masking process prior to chemical processing and/or paint.

1.2. Definitions

Term	Definition
FTS	Finish Technique Sheet
IAW	In accordance with
Maskant	Materials used to cover sections of the parts to omit finish.
Part Masking	The act of applying material to cover the section of the part to omit finish.
Masking Finish Technique Sheets (FTS)	Pictorial, graphical illustration used to properly outline process requirement.
Media Engineering	Engineering supplied by the customer and/or Skills Inc. that details the specific masking requirements. (i.e. Drawings, Part List (PL), Pictured Sheet Data List (PSDL), Photo Contact Master (PCM), Masking technique sheets and templates.
Masking Template	A piece of metal, acrylic sheet, plastic or other material that is pre-engraved or cut to shape. These templates are often used in Masking, Part marking or other processes.
Pre-cut masking	Laser engraved masking material usually 8992L film.
Secondary Container	Any container other than the container received from the manufacturer that a material has been placed in.
Skills Drawing	Simplified version of 2D and/or 3D CAD drawing used to detail process requirement (Skills internal drawing).
P.O.	Customer Purchase Order/Contract
Masking tools	Any tools used in masking. Caliper, Masking templates, pre-cut masking, rulers, pliers, plugs, etc.
WO	Work Order

1.3. References

Type of Document	Document Number	Document Title
Skills Inc.	SKA_2050	Control and Labeling of Shelf Life Materials
Skills Inc.	SKQ_3040	Control of Monitoring and Measuring Equipment
Skills Inc.	SKQ_3180	Control of Documents and Records
Skills Inc.	SKQ_3235	Work Record Requirements
Skills Inc.	SKQ_3245	Foreign Object Debris-Damage (FOD) Prevention
Skills Inc.	SKQ_3425	Control of Nonconforming Outputs

Uncontrolled When Printed



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Skills Inc.	Standard Work	AAM Review for Masking
Skills Inc.	Standard Work	Masking with Adcoat
Skills Inc.	Standard Work	Masking with 8992L
Skills Inc.	Standard Work	Part Handling
Skills Inc.	Standard Work	Part Verification
Boeing	PS20722	Hole Masking for Enhanced Process
		Controlled Metallic Parts

2. RESPONSIBILITIES

2.1. Masking Department Supervisor/Manager is responsible for:

- Maintaining this procedure to requirements;
Verify masking material is compatible with hardware, process conditions, can be removable, leaves no residue, and withstands temperatures when exposed during processing. (i.e. masking material is per BAC5034-4)
- Ensure employees are trained to requirements and have the materials/tools needed to perform their job duties.

2.2. Masking Personnel is responsible for:

- AAM verification on previous work order operations to ensure they are completed and legible, prior to performing their operation;
- Complete work order operation in accordance to Customer and Skills Inc. requirements;
- Report suspected and/or identified non-conforming material/parts or process to Lead, Supervisor, Manager, or Quality Assurance.

3. TRAINING

Only certified personnel can perform PS20722, Hole Masking for Enhanced Process Controlled Metallic Parts.

4. MATERIALS/ MASKANT

- 8992L Tape
- Ad Coat; AC-828-T Tan
- Gloves, Lint Free, Cotton
- Solid Pull Plugs IAW 9M1084
- Tape / Plugs / Paper
- Tools used in this area have potential to create FOD in this or other processes. To reduce the risk of FOD, all tools used in this area should be accounted for. If you identify a tool used in this process is missing inform your Lead/Supervisor and/or Quality (Inspection) immediately, to ensure tools are not traveling with product.



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5. PROCEDURE

5.1. General Requirements

- 5.1.1. Verify the previous (2) work order operations IAW AAM Review for Masking Standard Work; complete work order operation IAW SKQ_3235, Work Record Requirements.
- 5.1.2. Handle parts IAW Part Handling Standard Work
- 5.1.3. Secondary containers must be properly labeled IAW SKA_2050, Control & Labeling of Shelf Life Materials. (I.e. Adcoat, Alcohol, MEK, MPK, and sanitizer containers/bottles)
- 5.1.4. Keep work areas clean as you go, FOD/FOD free through shift IAW SKQ_3425, Foreign Object Debris/Damage (FOD) Prevention. Wear clean, lint free, powder free gloves when handling parts for masking to prevent handling contamination.

5.2. Order of Precedence

In the event of a conflict in the masking requirements where precedence cannot be established through the customer purchase order the following document hierarchy and considerations shall apply:

- a) Contract \ Purchase Order
- b) Model-Based Definitions (MBD)

Note: When MBD is noted in the engineering section of the customer purchase order, use Skills Inc. Generated Drawings for the masking requirements.

- c) Engineering Drawings
- d) Tech sheet & Skills Drawings
- e) None of the above, notify your supervisor or lead before proceeding with masking.

5.3. Engineering Review

Job should contain a Work Order, PL, PSDL and Drawing:

- Part List details finish requirement such as chem treat, primer and topcoat;
 - Picture sheet data list (PSDL) provides detail description of flag notes;
 - Drawing identifies masking or other process requirements;
 - FTS provides visual illustration of masking area;
 - P.O. outlines the processing details by customer, attached to the back of the WO;
 - Masking Operation in Work Order provide short instruction of masking process.
- 5.3.1. First verify part's documents mainly customer drawing, FTS, Skills drawing and/or LWG, PSDL, PL and PO are present and correct revision to work order.
 - 5.3.2. Confirm part's shape matches the drawings IAW Part Verification Standard Work.
- If the necessary documents are missing and/or the part does not conform, stop and report to Supervisor/Quality Assurance.



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5.4. Masking Area/Location

Masking instruction within the WO operation summarizes the requirements for masking.

5.4.1. FTS/Skills drawing/Masking Manual

Finish Tech sheet (FTS) provides visual appearance of part/masked part, masking types & location, material, and procedure. Skills drawing provides 2D and 3D views with masking information. Some of the part has masking manual due to its complexity.

Ensure masking manual are up to date with drawings.

If the WO lists Skills tech sheets or drawings they are located in the following locations. Contact your Supervisor, if you are unable to access them:

Plant 1 Drawings:

- Regular: G:\Configuration Management\Tech Sheets – Released
- ITAR: G:\Configuration Management\Tech Sheets ITAR – Released

Skills Drawing generated in Plant 2:

- Regular S:\P2\Configuration Management\Skills Generated Drawings Library\Skills Generated Drawings
- ITAR: S:\P2\Configuration Management\Skills Generated Drawings Library\Skills Generated Drawings ITAR

5.4.2. Customer drawing/PL/PSDL

Part list called Customer drawings are often the authority. Verify masking details in the drawing and refer PSDL as needed for flag note callout.

5.4.3. Purchase Order

In some cases, if the masking is simple enough such as plugging holes, purchase order indicates holes to be mask for specific processes. Cross check this information with PL, drawing & PSDL.

5.4.4. MBDs/LWG

Model-Based Definitions (MBD) is a 3D model which effectively contain all the data needed to define a product. This model contains all the requirements including finish and masking details. LWG is a copy of MBD. When masking is simpler, the job may have LWG as a drawing reference.

It is required to cross-check masking instruction with engineering listed.

5.5. Cleaning prior to Masking

Surface shall be cleaned and free of all contaminates prior to masking.



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5.6. Masking Methods/Types

Identify masking location and determine appropriate masking method and type. If a part is new and masking is first run, consult with masking supervisor before masking.

Special masking: Unless otherwise specified in customer purchase order or engineering documents, paint will not be applied to the inside of ground stud holes, close tolerance holes, threaded holes and bushing bores (BSS 8.2.1 C).

When masking to a dimensional tolerance, it is recommended to choose the middle tolerance range. This reduces the rejections for the masking being out of tolerance.

When overspray is allowed, masking may not be required; however, it depends on shape and size, way it's being set up. Some of them require a strip of tape with "Do Not Paint" written on it, mask accordingly.

5.6.1. Primer/Topcoat Masking

- Use tape/plugs as required to mask parts;
- Masking templates are located in respective masking bags. If they are larger, they are stored in drawers or hung corner of the wall in masking area. These templates are marked with part numbers for easier access.
- Pre-cut masking is arranged in numerical order by part number, find correct masking template in drawer. If pre-cut masking is out, order more from drafting room. Return unused masking back to correct location. Pre-cut masking is made to the tolerance. Verify as needed.

If available within the WO operation, circle type of maskant used – Tape, Plugs, Liquid.

Note: Precut masking are designed each for specific part numbers, do not use one for the other.

5.6.2. Electrical Bond

Requirement for electrical bond: Avoid bubbles and blisters under the tape or around loose plugs which can cause chemical leakage. Use squeegee and/or scotch brite pads to gently press down the tape for better adhesion.

While applying Adcoat, apply multiple layers, making it thicker. Thicker layer will peel off efficiently later.

- Using 8992L tape: Mask IAW standard work "Masking with 9882L"
- Electrical Bond using Adcoat: Mask IAW standard work "Masking with Adcoat"

If available within the WO operation, circle type of maskant used – Tape, Plugs, Liquid

5.6.3. PS20722, Hole Masking for Enhanced Process Controlled Metallic Parts

Only certified personnel shall perform and inspect operations in accordance with engineering and specification using solid pull plugs IAW 9M1084.



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6. Records

- SKA_2110, Work Order Traveler

7. Revision History

7.1. Add revision history-only keep last three revision changes

Revision	Changes Made
Rev A	Initial release
Rev B	Updated format to procedure, title change from Chemical and Paint Masking to Finish Masking; complete re-write to current process.
Rev C	Updated section 4 to add tool control per FOD procedure.